Tertiary Infotech Academy Pte Ltd

# Lab 15 — Build a Clean-to-Risk Pipeline with pipe

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## Scenario

A credit-risk team wants the DataFrame transformations to read like an auditable sequence rather than one large notebook cell.

## Goal

A chain of named clean, enrich, score and summarise functions connected with pipe().

## Tools

Pandas, CSV, unittest

## Step-by-step

**1.** Load loan_pipeline.csv and preserve a raw copy for comparison.

**2.** Write clean_data(df) without mutating its input.

**3.** Write add_features(df) for affordability and utilisation ratios.

**4.** Write score_risk(df) using vectorised conditions or apply.

```
python starter.py
```

**5.** Chain the functions with pipe and export results.

```
python solution.py
```

**6.** Run tests for immutability, boundaries and row counts.

```
python -m unittest test_solution.py
```

## Acceptance criteria

Each stage can be tested alone, the raw DataFrame is unchanged, and the final risk-band counts reconcile to total rows.

## Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.